📖 DEFINITIONS & GLOSSARY

Comprehensive Reference: 33 Technical Terms Defined with Everyday Analogies

PART 1: NETWORKING & CORE SECURITY FUNDAMENTALS

1. Cybersecurity

The practice of protecting systems, networks, and programs from digital attacks and unauthorized access.

💡 *Like locks, guards, and alarm systems protecting a bank.*

2. Confidentiality (CIA)

Ensuring information is accessible only to those authorized to have access.

💡 *Like sealing a letter inside a tamper-proof envelope.*

3. Integrity (CIA)

Ensuring information remains accurate, complete, and uncorrupted by unauthorized modification.

💡 *Like a wax seal verifying a contract hasn't been rewritten.*

4. Availability (CIA)

Ensuring systems and data are reliably accessible to authorized users when needed.

💡 *Like a 24/7 store keeping its front doors unlocked and staffed.*

5. Client and Server

Client is the device requesting data; Server is the centralized host providing the service.

💡 *A restaurant customer (Client) placing an order with the kitchen (Server).*

6. IP Address and Port

IP identifies the device on the network; Port identifies the specific software application listening.

💡 *IP is the building street address; Port is the specific apartment number.*

7. Network Packet

A formatted unit of data routed between origin and destination over the Internet.

💡 *A single postal postcard with a recipient address and small note.*

8. Protocol

A formal set of rules governing how devices exchange information across networks.

💡 *A shared language (e.g. English) ensuring two diplomats understand each other.*

9. ICMP Protocol

Internet Control Message Protocol (Layer 3) used for network error reporting and diagnostic pings.

💡 *A walkie-talkie radio check ("Can you hear me? Over.").*

10. Echo Request & Echo Reply

ICMP Type 8 message sent to ping a host, and the ICMP Type 0 reply confirming it is alive.

💡 *Clapping hands in an empty hall and listening for the acoustic echo.*

11. TCP and UDP

TCP provides reliable, ordered connection handshakes; UDP is connectionless, fast, and best-effort.

💡 *TCP is certified registered mail; UDP is tossing flyers out of an airplane.*

12. HTTP and HTTPS

HyperText Transfer Protocol for web pages; HTTPS encrypts transmission via TLS/SSL.

💡 *HTTP is speaking loudly in public; HTTPS is whispering in secret code.*

PART 2: ATTACK TYPES & MECHANISMS

13. Flood Attack

An offensive tactic transmitting excessive volume to starve a target's bandwidth or CPU processing.

💡 *1,000 pizza delivery drivers arriving at your doorstep simultaneously.*

14. DoS vs. DDoS

Denial-of-Service uses one computer; Distributed DoS commands a botnet of thousands of machines.

💡 *One protestor blocking a revolving door vs an organized flash mob.*

15. SYN Flood

Layer 4 attack initiating thousands of TCP handshakes without completing them, filling connection memory.

💡 *Dialing a restaurant phone, asking for a table, and leaving the phone off the hook.*

16. UDP Flood

Sending high-speed UDP packets to random ports, forcing target to look up listeners and send ICMP errors.

💡 *Mailing packages to nonexistent rooms in a hotel, exhausting the front desk.*

PART 3: PERFORMANCE METRICS & TELEMETRY

17. HTTP Flood

Layer 7 flood sending complex web requests (search queries, database downloads) to overload web servers.

💡 *Ordering the most complicated custom cocktail at a bar 500 times in a row.*

18. PPS (Packets Per Second)

The count of network packets passing through an interface in one second.

💡 *How many individual envelopes drop through your mailbox slot per second.*

19. Bandwidth

The maximum data transfer capacity of a network communication channel (e.g., 1 Gbps).

💡 *The width of a highway lane allowing cars to travel side-by-side.*

20. Latency (RTT)

The time delay taken for data to travel from sender to destination and return back.

💡 *How long you wait between asking a question and getting an answer.*

21. Packet Loss

Percentage of transmitted packets that fail to reach their destination due to full buffers.

💡 *Dropping packages out of an overloaded mail truck.*

22. CPU & Memory Utilization

Percentage of compute processor cycles and RAM state currently occupied by running tasks.

💡 *How much of a chef's attention and counter space is occupied cooking.*

23. Traffic Baseline

The typical, expected pattern of network traffic measured during normal day-to-day operations.

💡 *Knowing average morning commute traffic so you can spot an accident delay.*

24. False Positive

When legitimate, benign traffic is mistakenly flagged as malicious and blocked.

💡 *A car alarm screaming when a stray cat jumps on the hood.*

PART 4: DEFENSE, MITIGATION & CRYPTOGRAPHY

25. Firewall

Network security device monitoring incoming/outgoing traffic based on security policies.

💡 *A security checkpoint at a gated residential community.*

26. Rate Limiting

A defensive policy capping how many requests a user or IP can submit within a time interval.

💡 *An amusement park ride allowing only 20 riders every 5 minutes.*

27. Traffic Filtering

Examining packet headers and dropping any packets matching known malicious signatures or ports.

💡 *A bouncer refusing entry to anyone not on the verified VIP guest list.*

28. Traffic Scrubbing

Routing incoming network traffic through high-capacity filtering centers that strip attack traffic.

💡 *Passing dirty river water through industrial purifiers before delivery to taps.*

29. Load Balancing

Distributing network traffic across multiple backend servers to prevent any single server from overloading.

💡 *Opening 10 cashier lanes at a supermarket instead of funneling everyone to one.*

30. Mitigation

The process of reducing the severity or stopping the impact of an active cyber attack.

💡 *Firefighters deploying hoses to contain and extinguish a building fire.*

31. Encryption

Encoding plaintext into ciphertext so that only authorized parties with the key can read it.

💡 *Writing a secret message in a substitution code only your best friend knows.*

32. Hashing & Digital Signature

Hashing produces a fixed-size mathematical fingerprint; Digital signatures verify authenticity.

💡 *A tamper-evident hologram seal and royal signature on an official decree.*